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Influence of initial configuration in growing random graph models

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0.1 Introduction

The aim of this thesis is to give a brief introduction to the preferential attachment model, and to study the evolution of the preferential attachment graphs, from a fixed tree. Preferential attachment models are used to model complex networks such as the world wide web. The question we are interested in, is whether the early stage of a preferential attachment tree determine its asymptotical behaviour to some extent. This problem is the same as conditioning on the event that a random tree first evolves into a fixed tree S, and investigating its further evolution. This tree S is the so called seed of the preferential attachment tree. While investigating this phenomenon, first we need to define preferential attachment graphs. We will do so in Chapter 1, where we define various variations of the model as well.

In Chapter 2 we introduce the local weak limit of random graphs, a method often used when investigating random graphs, and also define the limiting object of preferential attachment trees.

In Chapter 3 we define formally the problem of investigating the evolution of preferential attachment trees from a fixed tree. While doing so we give a brief introduction to some quantities from information theory, and prove the result of Bubeck et al. ([BMR15]), that the seed has no effect on the local weak limit of the preferential attachment model . However, we will see that there is a function that can measure the effect of the seed on the resulting graph([BMR15]). This function is the limit of the total variation distance of the two preferential attachment graphs starting from seeds S and T.

In Chapter 4 we give a brief proof by Curien et al. ([CDKM15]) to the proposition that the limit of the total variation distance of two preferential attachment graphs starting from seeds S and T is a metric on trees with at least 3 edges. After this we give an introduction to uniform attachment trees, and state a similar theorem proved by Bubeck et al. [BEMR17], namely that that the limit of the total variation distance of two uniform attachment graphs starting from seeds S and T is a metric on trees with at least 3 edges.

In the fifth chapter we feature some algorithmic results for finding the seed of the uniform attachment tree, and discuss some open problems regarding this area.

Chapter 1

Preferential Attachment Model

In this chapter we wish to give a brief summary of the preferential attachment model and its properties. In this thesis we focus mostly on preferential attachment trees, but some results apply to a wider selection of preferential attachment graphs, therefore we define them in general.

Preferential attachment graphs and their real life applications were first investigated by Yule in 1925. In 1999, Albert-László Barabási and Réka Albert published [BA99], which is considered as the first modern approach of the topic. They were looking for a network model that has similar characteristics to such real life networks as the World Wide Web or genetic networks. One such characteristic was the network's vertex connectivities following a scale-free power law distribution, which was found to be the consequence of new vertices being added to the network, and preferentially connecting to already well-connected parts of the network. [VDH17]

1.1 Preferential attachment trees

We define the preferential attachment tree as it can be found in [Mó11], and according to the terminology of [BMR15]. When looking at the definition of the preferential attachment graphs, we give another definition of the trees, which can be found in [VDH17].

Definition 1.1.1 (Preferential attachment tree) *The preferential attachment tree, PA is a random recursive tree on labeled vertices. The following notations will be used:*

1. *the vertices of the tree after n steps are $V_n = \{0, 1, 2, \ldots, n\}$*

2. the degree of vertex i in the graph after n steps is $d_n(i)$.

$PA(1)$ is the vertices $0, 1$ connected by a single edge. Given $PA(n)$, $PA(n+1)$ is formed by adding a new vertex, $n+1$ to $PA(n)$, and connecting it to the previous vertices with one edge, which is chosen according to the following distribution:

$$\mathbb{P}(v_{n+1} \to v_i | PA(n)) = \frac{d_n(i)}{2n}.$$

We move on from the definition by stating and proving two theorems about the average and maximum degree of preferential attachment trees. These proofs rely on martingale techniques, and can be read in their full extent in [Mó11].The reason we present this proof is due to the martingale techniques presented in the proof. One of the main results that we discuss in this thesis is Theorem 1 in [CDKM15], and the method used to prove that theorem is more technical, but the proof is very similar in nature to the proof of the following theorem.

Theorem 1.1.2 ([Mó11] 11.1) *Let $X_n(d)$ be the number of vertices with degree d in a preferential attachment tree after n steps. For all $d \in \mathbb{N}$ the following limit holds*

$$\lim_{n\to\infty} \frac{X_n(d)}{n-1} = \frac{4}{d(d+1)(d+2)}$$

almost surely.

Proof. Let $\mathcal{F}_n$ denote the σ-algebra after n steps. We first show that the theorem holds for $d = 1$.
We start by calculating the conditional expectation of $X_n(1)$: In order to understand $\mathbb{E}(X_{n+1}(1)|\mathcal{F}_n)$ we need to first understand how $X_n(1)$ impacts $X_{n+1}(1)$. The change in the number of vertices with degree 1 after one step is the following: the new vertex added to the graph will have degree one, and – due to the preferential attachment dynamics – with probability $\dfrac{X_n(1)}{2n}$ we will lose one vertex with degree one. This written out with conditioning on $\mathcal{F}_n$ is the following:

$$\mathbb{E}(X_{n+1}(1)|\mathcal{F}_n) = X_n(1) - \frac{X_n(1)}{2n} + 1 = \left(1 - \frac{1}{2n}\right) X_n(1) + 1.$$

Now let $c_n = \dfrac{\Gamma(n)}{\Gamma(n - \frac{1}{2})}$, since $\dfrac{c_n}{c_{n+1}} = 1 - \dfrac{1}{2n}$ we get that

$$\mathbb{E}(X_{n+1}(1)|\mathcal{F}_n) = \frac{c_n}{c_{n+1}} X_n(1) + 1. \tag{1.1}$$

For $Y_n = c_n X_n(1)$, after multiplying (1.1) by c_{n+1} we get:

$$\mathbb{E}(Y_{n+1}|\mathcal{F}_n) = Y_n + c_{n+1},$$

which means that $(Y_n, \mathcal{F}_n)$ is a submartingale.
We shall use the Doob decomposition theorem, which would give us $Y_n = A_n + Q_n$, where Q_n is a martingale, and A_n is predictable. We shall inspect A_n and Q_n separately.
For A_n we get that,

$$A_n = \mathbb{E}(Y_1) + \sum_{i=2}^{n} (\mathbb{E}(Y_i|\mathcal{F}_{i-1}) - Y_{i-1}) = 2c_1 + \sum_{i=2}^{n} c_i.$$

We will use the notation $a_n \sim b_n$, if $\lim_{n\to\infty} \frac{a_n}{b_n} = 1$. Since $c_n \sim \sqrt{n}$, we get that $A_n \sim \frac{2}{3} n^{\frac{3}{2}}$.
Moving on to the martingale, we want to calculate its canonical growth process, for which we acquire the following:

$$\begin{aligned} B_n &= \sum_{i=1}^{n} D^2(Y_i|\mathcal{F}_{i-1}) = \sum_{i=1}^{n} c_i^2 D^2(X_i(1)|\mathcal{F}_{i-1}) = \sum_{i=1}^{n} c_i^2 D^2(X_i(1) - X_{i-1}(1)|\mathcal{F}_{i-1}) \leq \\ &\leq \sum_{i=1}^{n} c_i^2 \mathbb{E}^2((X_i(1) - X_{i-1}(1))^2|\mathcal{F}_{i-1}) \leq \sum_{i=1}^{n} c_i^2 = O(n^2). \end{aligned}$$

Exploiting Theorem 1.1.4 we get that $Q_n = o(B_n^{\frac{1}{2}} \log(B_n)) = o(A_n)$ on the event $\{B_n \to \infty\}$, which remains true when B_n also converges.
From this we get that $Y_n \sim A_n$, and since $c_n \sim n^{\frac{1}{2}}$, we get that $X_n(1) \sim \frac{2}{3} n$.
Now we proceed by conducting induction; let us assume that $X_n(d-1) \sim \dfrac{4n}{(d-1)d(d+1)}$.
We obtain the conditional property similarly to the case $d = 1$, since the number of vertices with degree d can change after adding a new vertex to the graph by either a vertex with $d-1$ neighbors acquiring a new neighbor and becoming a vertex with d neighbors, or a vertex with d neighbors getting a new neighbor, thus not being a vertex with d neighbors anymore. By conditioning on $\mathcal{F}_n$, and exploiting the preferential attachment dynamics we get:

$$\begin{aligned} \mathbb{E}(X_{n+1}(d)|\mathcal{F}_n) &= X_n(d) - \frac{dX_n(d)}{2n} + \frac{(d-1)X_n(d-1)}{2n} = \\ &= \left(1 - \frac{d}{2n}\right) X_n(d) + \frac{(d-1)X_n(d-1)}{2n}, \qquad n \geq d. \end{aligned}$$

We shall use a similar constant as in the $d = 1$ case. Let

$$c_n = \frac{\Gamma(n)}{\Gamma(n - \frac{d}{2})} \sim n^{d/2}, \tag{1.2}$$

and $Y_n = c_n X_n(d)$. This is a submartingale as well, with a Doob decomposition $Y_n = A_n + Q_n$. For A_n we get

$$A_n = \mathbb{E}(Y_d|\mathcal{F}_{d-1}) + \sum_{i=d+1}^{n} (\mathbb{E}(Y_i|\mathcal{F}_{i-1}) - Y_i) \sim \sum_{i=d+1}^{n} c_i \frac{d-1}{2} \frac{X_{i-1}(d-1)}{i-1} \sim$$
$$\sim \frac{2}{d(d+1)} \sum_{i=d+1}^{n} i^{d/2} \sim \frac{4}{d(d+1)(d+2)} n^{d/2+1}.$$

Q_n is a square integrable martingale and its canonical growth process can be bounded the same way as for $d = 1$. We get that

$$B_n = \sum_{i=1}^{n} D^2(Y_i|\mathcal{F}_{i-1}) = O(n^{d+1}).$$

By using Theorem 1.1.4 again, on the event $B_n \to \infty$ we have $Q_n = o(\sqrt{B_n}\log(B_n)) = o(A_n)$, and if B_n converges $Q_n = o(A_n)$ remains true, which yields $Y_n \sim A_n$. Furthermore, after exploiting (1.2), we get

$$X_n \sim n^{-d/2} A_n \sim \frac{4}{d(d+1)(d+2)},$$

which concludes the proof. □

Now we move on to the maximum degree. The importance of this theorem will be emphasized later, when we are investigating the effect of the seed. We will find that the seed has an effect on the limiting distribution of the maximum degree.

Theorem 1.1.3 ([Mó11] 11.2) *For some random variable $D_{\max} > 0$ we have*

$$\frac{M_n}{\sqrt{n}} \to D_{\max}$$

almost surely as $n \to \infty$, where M_n is the maximum degree of the preferential attachment graph after n steps.

Before moving on to proving the theorem we will state another theorem which will be needed later.

Theorem 1.1.4 ([Mó11] 5.5) *Let $(X_n, \mathcal{F}_n)$ be a square integrable martingale, A_n its predictable growth process and $f : \mathbb{R}^+ \to \mathbb{R}^+$ a monotone increasing function such that*

$$\lim_{x\to\infty} f(x) = \infty \quad \textit{and} \quad \int_0^\infty \frac{1}{(1+f(t))^2} dt < \infty$$

hold. Then on the event $\{A_\infty = \infty\}$ we have $X_n = O(f(A_n))$.

Proof of Theorem 1.1.3. The proof is similiar in nature to the one on the convergence of average degree. For every $k \in \mathbb{N}^+$ we define

$$q_n(k) = \frac{\Gamma(k)}{\Gamma(n+\frac{k}{2})} \sim n^{-k/2}.$$

These constants satisfy the following equation:

$$\frac{q_n(k)}{q_{n+1}(k)} = 1 + \frac{k}{2n}.$$

Let $Z_n = Z_n(i,k) = q_n(k)\binom{d_n(i)+k-1}{k}$, for $n \geq i$. We continue by showing that Z_n is a martingale with respect to the filtration of the preferential attachment tree.

$$\begin{aligned}
\mathbb{E}\left(\binom{d_{n+1}(i)+k-1}{k}\Big|\mathcal{F}_n\right) =& \left(1-\frac{d_n(i)}{2n}\right)\binom{d_n(i)+k-1}{k} + \frac{d_n(i)}{2n}\binom{d_n(i)+k}{k} = \\
=& \binom{d_n(i)+k-1}{k}\left(1-\frac{d_n(i)}{2n}+\frac{d_n(i)+k}{2n}\right) = \\
=& \binom{d_n(i)+k-1}{k}\frac{q_n(k)}{q_{n+1}(k)},
\end{aligned}$$

since in the first line we exploited that $d_{n+1}(i)$ is $d_n(i)$, given that no edge was connected to it, or $d_n(i)+1$ given that an edge was connected to it.

Given that this is a non-negative martingale, it converges almost surely. If $k=1$ holds, we have that

$$Z_n(i,1) = q_n(1)d_n(i) \to \zeta_i$$

almost surely, and due to the properties of the gamma function we have $q_n(k) = q_n(1)^k$. Combining these yields $Z_n(i,k) \to \frac{\zeta_i^k}{k!}$.

We wish to continue by determining the moments of ζ_i.
Let p be a positive integer. We will show that $(Z_n(i,k))^p$ converges in L_p.

$$(Z_n(i,k))^p = (q_n(k))^p\binom{d_n(i)+k}{k-1}^p = \frac{(q_n(k))^p}{(k!)^p}\left[\prod_{j=0}^{k-1}\left(d_n(i)+j\right)\right]^p \leq \tag{1.3}$$

$$\leq \frac{q_n(k))^p}{(k!)^p}\prod_{j=0}^{kp-1} d_n(i)+j = \frac{q_n(k))^p}{(k!)^p}\frac{(kp)!}{q_n(kp)}Z_n(i,kp). \tag{1.4}$$

The coefficient of $Z_n(i,kp)$ converges as n tends to infinity, which means the right-hand side remains bounded in L_1, which yields it remains bounded for all positive integers p,

thus it is convergent in L_p for every p.
This yields

$$\frac{\mathbb{E}(\zeta_i)^k}{k!} = \lim_{n\to\infty} Z_n(i,k) = \mathbb{E}(Z_i(i,k)) = q_i(k), i \leq 1. \tag{1.5}$$

We continue by calculating the distribution of ζ_0.
Since the distribution of $d_n(0)$ and $d_n(1)$ are identical, the same applies to ζ_0 and ζ_1, which means that

$$\mathbb{E}\zeta_0^k = \frac{\Gamma(k+1)}{\Gamma(\frac{k}{2}+1)}.$$

By exploiting the properties of the gamma function and the gamma distribution we get that the moments of ζ_0 are identical to the square roots of the moments of a gamma distribution. From this we only need that $\mathbb{P}(\zeta_0 > 0) = 1$.
Since $q_n(1)M_n = \max_i Z_n(i,1)$ is a submartingale, we get

$$\mathbb{E}(M_{n+1}q_{n+1}(1)|\mathcal{F}_n) \geq q_n(1)M_n.$$

Let $p \geq 3$. We have that

$$\mathbb{E}(q_n(1)M_n)^p = \mathbb{E}(\max_i Z_n(i,1))^p \leq \sum_{i=1}^{n} \mathbb{E}(Z_n(i,1))^p,$$

and according to equation (1.3), $Z_n(i,1)$ converges in L_p to ζ_i, and $\mathbb{E}(Z_n(i,1))^p$ is monotone increasing. Combining these we get

$$\mathbb{E}(q_n(1)M_n)^p \leq \sum_{i=0}^{n} \mathbb{E}\zeta_i^p < \sum_{i=0}^{\infty} \mathbb{E}\zeta_i^p = p! \sum_{i=0}^{\infty} q_i(p) < 0,$$

which means that $M_n(1)q_n(1)$ converges almost surely and in L_p. In addition, since $q_n(1) \sim \frac{1}{\sqrt{n}}$, we have that $\frac{M_n}{\sqrt{n}} \to D_{\max}$ almost surely. □

This result will be revisited in Chapter 3, where we will calculate the limiting distribution more explicitly.

Using similar techniques as before, we can get a limit theorem for the degree of a fixed vertex. We postpone the proof of this theorem to the next subsection, because we will prove it in a more general way.

Theorem 1.1.5 ([VDH17] Theorem 8.2 and Exercise 8.14) *Let $d_n(i)$ be the degree of the vertex i in $PA(n)$ after n steps. Then for some random variable D_i we have for all i as $n \to \infty$:*

$$\frac{d_n(i)}{\sqrt{n}} \xrightarrow{a.s.} D_i,$$

and

$$\mathbb{E}(d_n(i)) = \frac{\Gamma(n+1/2)\Gamma(i)}{\Gamma(i+1/2)\Gamma(n).}$$

It can be also shown that

$$D_{\max} = \max_{i\in\mathbb{N}} D_i.$$

This will be exploited later, when we want to deduce bounds for the maximum degree of the preferential attachment tree starting from an arbitrary seed.

1.2 Different variations of preferential attachment graphs

In this section we will define several modifications of the preferential attachment model. Throughout this chapter we will define models as it can be found in [VDH17]. As there are several modifications to the model, we will try to emphasize under which conditions the definitions are equivalent.

1.2.1 Affine preferential attachment graphs

We start by defining the preferential attachment graph, which connects every new vertex with a single edge. From this we can generalize it for a model where a new vertex can have multiple edges.

Definition 1.2.1 *Let $\delta \geq -1$, and $PA_1^{1,\delta}$ be a graph that consists of a single vertex with a self-loop. Let the vertices of $PA_n^{1,\delta}$ be $v_1, v_2, \dots, v_n$, and the degree of a vertex $d_n(i)$, with the convention that a self-loop increases the degree of a vertex by two. From $PA_n^{1,\delta}$ we get $PA_{n+1}^{1,\delta}$, by adding a new vertex and an edge to the graph, which connects v_{n+1} to another point with the following probability:*

$$\mathbb{P}(v_{n+1} \to v_i | PA_n^{1,\delta}) = \begin{cases} \dfrac{1+\delta}{n(2+\delta)+1+\delta}, & \textit{if } i = n+1 \\ \dfrac{d_n(i)+\delta}{n(2+\delta)+1+\delta}, & \textit{if } i \in [n]. \end{cases}$$

The main difference of this definition and the one in the first section is that self-loops are allowed here. The reason for that comes from the initialization of the model. If we start from one vertex, and self-loops are not allowed, the new vertex would connect its edge with zero probability to any other vertex, thus the model would not be well-defined.

Another slight difference is the δ parameter introduced in this definition. It has an impact on the maximal degree and the fixed degree distribution as well, however we are not interested in the proof of these theorems.

Before discussing the aforementioned differences between the two models, we wish to define the preferential attachment graphs, where a new vertex can have multiple edges.

Definition 1.2.2 (Preferential attachment graph) *Let $\delta \geq -m$, where m is the number of new edges added to the graph after adding a new vertex.*
The graph for $m > 1$ is defined in terms of the model $m = 1$ as follows.
Let $\delta \geq -m$, and $PA_{mn}^{1,\delta/m}$ a preferential attachment graph after mn steps, with vertices $v_1^1, v_2^1, ...v_{mn}^1$. From this we get $PA_n^{m,\delta}$ by collapsing the vertices of $PA_{mn}^{1,\delta/m}$, namely let $v_{k+1}^1, v_{k+2}^1, ...v_{2k}^1$ be vertices in $PA_{mn}^{1,\delta/m}$ collapsed into $v_k \in PA_n^{m,\delta}$, and every edge that runs between v_j^1 and v_l^1 in $PA_{mn}^{1,\delta/m}$ be a self-loop, if l and j correspond to the same vertex in $PA_n^{m,\delta}$, and an edge between two vertices of $PA_n^{m,\delta}$ if v_l and v_j correspond to different vertices of $PA_n^{m,\delta}$.

It is important to add to the above definition that when we add a new vertex to $PA_n^{m,\delta}$, we connect its m edges one by one, and after connecting each edge we update the degrees of the graph. This is called intermediate updating of the degrees. There are other methods to define how to connect the m edges but during this thesis we will only consider this one.

Differences of the two definitions

The main difference between the two definitions is the existence of self-loops in the second one, and the fact that a graph according to the latter definition might be disconnected.

However most theorems stated for a model based on either definition will have an almost completely analogous version for a model based on the other definition. The heuristic explanation for this is the fact that the probability of self-loops appearing tends to zero, and therefore can be ignored.

We are not going to prove the analogous version of Theorem [1.1.4] and Theorem [1.1.3], since it involves the same steps as we did throughout this chapter, and it can be found in [VDH17], but we will state Theorem [1.1.5] using the terminology introduced in this subsection, and prove it as well.
which has slightly more impact on these theorems, as it effects the limiting distribution. However, we will not address these differences, since it is not relevant to the topic of this thesis.

Theorem 1.2.3 (Degree of fixed vertices, Theorem 8.2 [VDH17]) *Let $m = 1$ and $\delta > -1$. Then $\frac{d_n(i)}{n^{1/2+\delta}}$ converges almost surely to a random variable D_i, and*

$$\mathbb{E}(d_n(i) + \delta) = (1 + \delta)\frac{\Gamma(n + 1)\Gamma(i - 1/(2 + \delta))}{\Gamma(n + \frac{1+\delta}{2+\delta})\Gamma(i)}.$$

Proof. Let $n \geq i$. We start by calculating the conditional expectation $\mathbb{E}(d_{n+1}(i)+\delta|d_n(i))$.

$$\begin{aligned}\mathbb{E}(d_{n+1}(i) + \delta|d_n(i)) =&d_n(i) + \delta + \mathbb{E}(d_{n+1}(i) - d_n(i)|d_n(i))\\ =&d_n(i) + \delta + \frac{d_n(i) + \delta}{(2 + \delta)n + 1 + \delta} =\\ =&(d_n(i) + \delta)\frac{(2 + \delta)n + 2 + \delta}{(2 + \delta)n + 1 + \delta} =\\ =&(d_n(i) + \delta)\frac{(2 + \delta)(n + 1)}{(2 + \delta)n + 1 + \delta}.\end{aligned}$$

Here we used the fact that $d_n(i)$ forms an inhomogeneous Markov chain. We continue by defining the following random variable:

$$X_i(n) = \frac{d_n(i) + \delta}{a + \delta}\prod_{s=i-1}^{t-1}\frac{(2 + \delta)s + 1 + \delta}{(2 + \delta)(s + 1)}, \tag{1.6}$$

which is a non-negative martingale with mean 1. Since a non-negative martingale always converges almost surely, we get that there exists a limiting variable X_i, such that

$$X_i(n) \xrightarrow{a.s.} X_i.$$

We continue by calculating the limiting distribution of $\frac{d_n(i)}{n^{1/2+\delta}}$, which we start by calculating the product in (1.6). By evaluating the product we get

$$\prod_{s=i-1}^{t-1}\frac{(2 + \delta)s + 1 + \delta}{(2 + \delta)(s + 1)} = \prod_{s=i-1}^{t-1}\frac{s + \frac{1+\delta}{2+\delta}}{s + 1} = \frac{\Gamma(t + \frac{1+\delta}{2+\delta})\Gamma(i)}{\Gamma(t + 1)\Gamma(i - 1/(2 + \delta))},$$

which combined with the definition of $X_i(n)$ yields:

$$X_i(n) = \frac{d_n(i) + \delta}{1 + \delta}\frac{\Gamma(n + \frac{1+\delta}{2+\delta})\Gamma(i)}{\Gamma(n + 1)\Gamma(i - 1/(2 + \delta))}. \tag{1.7}$$

For a fixed α, using the Stirling formula it can be easily seen that

$$\frac{\Gamma(n + \alpha)}{\Gamma(n)} = n^\alpha(1 + O(1/n)).$$

Applying this with $\alpha = \frac{1+\delta}{2+\delta}$, we get the appropriate limiting behaviour, namely

$$\frac{d_n(i)}{n^{1/2+\delta}} = X_i(n)\frac{(2+\delta)\Gamma(i-1/(2+\delta))}{\Gamma(i)}(1+O(1/n)) \xrightarrow{a.s.} X_i\frac{(2+\delta)\Gamma(i-1/(2+\delta))}{\Gamma(i)},$$

and by choosing $\zeta_i = X_i\dfrac{(2+\delta)\Gamma(i-1/(2+\delta))}{\Gamma(i)}$ we are done with the proof of the limiting distribution.

Next we want to calculate the expected degree of a fixed vertex. We will rely on the fact that

$$\mathbb{E}(X_n) = \mathbb{E}(X_{n+1}),$$

since X_n is a martingale. Combining this and the fact that

$$\begin{aligned}\mathbb{E}(d_i(i)+\delta) =& 1+\delta+\frac{1+\delta}{(2+\delta)(i-1)+1+\delta} = (1+\delta)\frac{(2+\delta)(i-1)+2+\delta}{(2+\delta)(i-1)+1+\delta} = \\ =&(1+\delta)\frac{(2+\delta)i}{(2+\delta)(i-1)+1+\delta}.\end{aligned}$$

with (1.7), we get

$$\mathbb{E}(d_n(i)+\delta) = (1+\delta)\frac{\Gamma(n+1)\Gamma(i-1/(2+\delta))}{\Gamma(n+\frac{1+\delta}{2+\delta})\Gamma(i)},$$

which concludes the proof. □

1.2.2 Preferential attachment graphs with fitness

We will now define a further modification of the model, the one with fitness. This modification can be interpreted from the perspective that the vertices have another property that make them more suitable for establishing new connections. This can be modeled as a random variable, which either has a multiplicative or an additive effect on the probability that a new edge would get connected to a given vertex.

We will only give the definitions of these models, more can be read on multiplicative case in [DO14], and the additive case in [LO20].

Preferential attachment model with additive fitness

Definition 1.2.4 *Let $(F)_{i\in\mathbb{N}^+}$ be a sequence of i.i.d. copies of a random variable F, taking values in $(0,\infty)$. For any $n \in \mathbb{N}^+$, we define*

$$S_n = \sum_{i=1}^{n} F_i,$$

and let $n_0, m_0 \in \mathbb{N}^+$. We say that a sequence of random graphs $(G_n)_{n \geq n_0}$ is a preferential attachment model with additive fitness if G_n is a directed weighted graph on the vertex set $[n]$ with edges directed from larger to smaller indices with the following properties: We assume that G_{n_0} has m_0 edges and we assign $F_1, F_2, \ldots, F_{n_0}$ to vertices $1, 2 \ldots, n_0$ respectively.

To obtain G_{n+1} from G_n for some $n \geq n_0$ as follows: we add vertex $n+1$ to the vertex set, assign F_{n+1} to $n+1$, and add m edges to the graph connecting $n+1$ to some other vertices, such that it fulfills one of the following three rules:

- ***Preferential attachment model with random out fitness**: $m = 1$, and conditionally on G_n, we connect $n+1$ to each vertex in $[n]$ with probability*

$$\frac{d_n(i) + F_i}{m_0 + (n - n_0) + S_n},$$

  with the restriction that, conditionally on G_n, $(d_{n+1}(i) - d_n(i)), i \in [n]$ are pairwise non-positively correlated.

- ***Preferential attachment model with fitness and fixed degree**: To vertex $n+1$ we assign m half-edges, and conditionally on G_n we connect each half-edge independently to some $i \in [n]$, with probability*

$$\frac{d_n(i) + F_i}{m_0 + m(n - n_0) + S}.$$

- ***Preferential attachment model with fitness and updating degree**: To vertex $n+1$ we assign m half-edges and let $d_n^j(i)$ denote the in-degree of vertex i when $n+1$ has attached j of its half-edges. For $j = 1, 2, \ldots, m$ conditionally on the graph of size $n+1$ including the first $j-1$ half-edges of $n+1$, we connect the j-th half edge with probability*

$$d_n^j(i) + F_i \frac{d_n^j(i) + F_i}{m_0 + m(n - n_0) + j - 1 + S_n}.$$

Preferential attachment graphs with multiplicative fitness

Definition 1.2.5 *Let $(F_i)_{i \in \mathbb{N}^+}$ be a sequence of i.i.d. copies of a random variable F, taking values in $(0, \infty)$ with distribution μ, which is a compactly supported distribution on the Borel sets of $(0, \infty)$. We say that the sequence of $(G_n)_{n \in \mathbb{N}^+}$ random graphs is a preferential attachment model with multiplicative fitness if G_n consists of $[n]$ vertices, G_1 is a single* *vertex without edges,* $F_1, F_2, \ldots, F_n$ *are assigned to* $1, 2, \ldots, n$ *respectively, and* G_{n+1} *can be acquired from* G_n *by adding* $n+1$ *to the graph, and connecting it to the* i*-th vertex of* $[n]$ *with a directed edge pointing at the older vertex with probability*

$$F_i(1 + d_n(i)).$$

Here we note that this is not enough for a unique description of a network, since the number of edges at each step cannot be calculated, without giving some further restrictions for $F_i(1 + d_n(i))$*.*
Let $\Delta\mathcal{I}^i_{j-1} = (1 + d_j(i)) - (1 + d_{j-1}(i))$*,* λ *an arbitrary positive parameter, and define*

$$\overline{F}_n = \frac{1}{\lambda n} \sum_{i=1}^{n} F_i(1 + d_n(i)).$$

Under the following assumptions, the model is well-defined:

- $\mathbb{E}[\Delta\mathcal{I}^i_n | G_n] = \dfrac{F_i(1 + d_n(i))}{n\overline{F}_n}$,
- *there exists a constant* C*, such that* $Var(\Delta\mathcal{I}^i_n | G_n) \leq C\mathbb{E}[\Delta\mathcal{I}^i_n | G_n]$*,*
- *conditionally on* G_n*, for* $i \neq j$ $\Delta\mathcal{I}^i_n$ *and* $\Delta\mathcal{I}^j_n$ *are non-positively correlated.*

There are similar results on the distribution of the maximum degree and average degree of the preferential attachment graphs with multiplicative and additive fitness, as on the distribution of the maximum degree and average degree of the general model. These theorems and their proofs can be found in [LO20], and in [DO14].

Chapter 2

Local convergence of graphs and random graphs

In this chapter we wish to give a brief introduction to the local limit of random graphs, which is a concept that was first introduced by Benjamini and Schramm in [BS01] in 2001, and independently by Aldous and Steele in [AS04] in 2004. Intuitively the local limit of a graph is the object we would see from the perspective of a typical vertex of the graph. In order to formulate this intuitive object, we will first define the local limit of deterministic graphs and then generalize the notion to random graphs. There are multiple approaches to defining the local limit, we will be following [VDH21] while doing so. Later we will see, that the seed has no impact on the local weak limit of the preferential attachment tree.

2.1 The metric space of rooted graphs

In order to define convergence, we need to define a metric space first. The metric space needed will be the metric space of rooted graphs, therefore we start by defining the rooted graphs first.

Definition 2.1.1 (Rooted graphs) *Let G be a graph, and $(V(G), E(G))$ be the set of its vertices and edges. A pair of (G, o) is a rooted graph if $o \in V(G)$.*

Definition 2.1.2 (Locally finite graphs) *We say that a graph G is locally finite if each of its vertices has finite degree.*

Now we can move on to defining the neighborhoods of graphs.

Definition 2.1.3 *Let (G, o) be a rooted graph, and for $u, v \in V(G)$ the distance of u, v is $d_G(u, v)$, is the regular distance of (u, v) in a graph. We denote the neighborhood of the root up to r vertices by $B_r^G(o) = (V(B_r^G(o)), E(B_r^G(o), o))$, where*

$$V(B_r^G(o)) = \{u \in V(G) : d_G(o, u) \leq r\}$$
$$E(B_r^G(o)) = \{\{u, v\} \in E(G) : d_G(o, u) \leq r, d_G(o, v) \leq r\}.$$

With the neighborhoods defined, the next question is when we can say that two neighborhoods are the same. For this we define the isomorphism of neighborhoods.

Definition 2.1.4 (Graph isomorphism) *Let $G_1 = (V(G_1), E(G_1))$, $G_2 = (V(G_2), E(G_2))$ be two (finite or infinite) graphs. We say that G_1 and G_2 are isomorphic, and write $G_1 \simeq G_2$, if there exists a bijection $\phi : V(G_1) \to V(G_2)$ such that an edge $e = (u, v) \in E(G_1)$ exists if and only if $((\phi(u), \phi(v)) = \phi(e) \in E(G_2)$.*

We will define isomorphism between rooted graphs identically, with the restriction that ϕ needs to map a root into a root. Let (G_1, o_1) and (G_2, o_2) be two rooted graphs, with $G_i = (V(G_i, o_i))$, for $i = 1, 2$. We say that (G_1, o_1) and (G_2, o_2) are isomorphic, and write $(G_1, o_1) \simeq (G_2, o_2)$, if there exists $\phi : V(G_1) \to V(G_2)$ such that $\phi(o_1) = o_2$, and an edge $e = (u, v) \in E(G_1)$ exists precisely when $((\phi(u), \phi(v)) = \phi(e) \in E(G_2)$.

With these definitions the notion of two graphs being identical from the perspective of a vertex make sense, as for a fixed r if the rooted graphs from that said vertex are isomorphic, it means that they cannot be distinguished. We continue by defining a metric on the set of rooted graphs.

Definition 2.1.5 (Metric on rooted graphs) *Let (G_1, o_1), (G_2, o_2) be two rooted connected graphs. Let*

$$R^\star = \sup_{r \in \mathbb{N}} \{r : B_r^{G_1}(o_1) \simeq B_r^{G_2}(o_2)\}.$$

We define the distance of two rooted graphs as $d_{G_\star}((G_1, o_1), (G_2, o_2)) = \dfrac{1}{R^\star + 1}$. When $R^\star = \infty$, the two graphs are isomorphic, and their distance is zero.

We denote the space of the rooted graphs by $\mathcal{G}_\star$. It can be shown that the $(\mathcal{G}_\star, d_{\mathcal{G}_\star})$ is a separable metric space, with the finite rooted graphs being a countable dense subset. Now we can advance to defining the local limit of deterministic graphs.

2.1.1 Local limit of deterministic graphs

Definition 2.1.6 *Let $G_n = (V(G_n), E(G_n))$ be a sequence of finite graphs, and (G_n, o_n) be the rooted graph acquired by picking a random vertex of G_n with uniform distribution, and restricting G_n to the connected component of o_n. We say that (G_n, o_n) converges in the local weak sense to (G, o), which is an element of $\mathcal{G}_\star$ having law μ, if for every bounded continuous function $h : \mathcal{G}_\star \to \mathbb{R}$*

$$\mathbb{E}(h(G_n, o_n)) \to \mathbb{E}_\mu(h(G, o)),$$

where the expectation on the right-hand side is with respect to (G, o) having law μ, and the expectation on the left-hand side is with respect to the sampling of o_n. We denote this convergence by

$$(G_n, o_n) \xrightarrow{d} (G, o).$$

This means that if we know the local limit of a graph, we know the limit of every bounded continuous function of "average" vertices. The problem can be the actual calculation of the local weak limit itself, since the local limit of simple deterministic graphs can be difficult to acquire. In order to emphasize this, we will give an easy example, where the local limit of a deterministic graph is a random graph.

Proposition 2.1.7 *Let G_n consist of $3n$ isolated vertices, and n copies of K_3, where K_3 is the complete graph on 3 vertices. The local weak limit of G_n is the random graph that is a single vertex with probability $\frac{1}{2}$, or K_3 with probability $\frac{1}{2}$.*

Proof. Let $h : \mathcal{G}_\star \to \mathbb{R}$ be an arbitrary bounded function, and let us start calculating $\mathbb{E}(h(G_n, o_n))$, with respect to the choice of o_n.

$$\mathbb{E}(h(G_n, o_n)) = \frac{1}{6n} \sum_{u \in V(G_n)} h(G_n, u) = \frac{1}{2} h(K_3, o) + \frac{1}{2} h(K_1, o),$$

which is exactly the desired random graph. □

This simple proposition showed the most elementary way of calculating the local weak limit of a random graph, however for more complex graphs, this method might fail. Next we are going to state two theorems on possible ways of determining the local limit. The proofs of these theorems can be found in [VDH21], and will be of key importance in understanding how to determine the local weak limit or random graphs.

Theorem 2.1.8 (Criterion for local convergence [VDH21] 2.6) *Let $((G_n, o_n))_{n\in\mathbb{N}^+}$ be a sequence of finite rooted graphs, and for every $H_\star \in \mathcal{G}_\star$ and every $r \in \mathbb{N}$*

$$p^{G_n}(H_\star) = \frac{1}{V(G_n)} \sum_{u\in V(G_n)} \mathbb{1}_{\{B_r^{G_n}(u)\simeq H_\star\}}.$$

(G_n, o_n) converges locally weakly to (G, o) having law μ precisely when

$$p^{G_n}(H_\star) \to \mu(B_r^G(o) \simeq H_\star).$$

With the help of this theorem, if we know enough about the graph G, and can understand the random variable $\mathbb{1}_{\{B_r^{G_n}(u)\simeq H_\star\}}$ well enough the limit can be acquired easily. However if G_n is not simple enough, the random variable $\mathbb{1}_{\{B_r^{G_n}(u)\simeq H_\star\}}$ can be difficult to understand. The next theorem is supposed to make that problem easier.

Theorem 2.1.9 ([VDH21] Theorem 2.8) *Let $((G_n, o_n)_{n\in\mathbb{N}^+}$ be a sequence of graphs, such that o_n is chosen uniformly at random, and let (G, o) be a random variable on $\mathcal{G}_\star$ with probability law μ, and $I \subset \mathcal{G}_\star$ a subset of the space of rooted graphs, such that*

$$\mu((G, o) \in I) = 1,$$

and let $I(r)$ be a subset of I, such that $I(r)$ consists of those elements of I, for which the distance of any vertex and the root is at most r. If

$$\mathbb{E}(p^{G_n}(H_\star)) \to \mu(B_r^G(o) \simeq H_\star),$$

for all $H_\star \in I(r)$, where

$$\mathbb{E}(p^{G_n}(H_\star)) = \frac{1}{V(G_n)} \sum_{u\in V(G_n)} \mathbb{P}(B_r^{G_n}(u)) \simeq H_\star),$$

then $(G_n, o_n) \xrightarrow{d} (G, o)$.

The significance of this theorem is that, if we can find a subset I, such that it contains the limit almost surely, we only need to calculate $\sum_{u\in V(G_n)} \mathbb{P}(B_r^{G_n}(u)) \simeq H_\star)$ for such $H_\star \in I(r)$. As an example to that we will prove the following corollary.

Proposition 2.1.10 *Let G_n be the path that consists of n vertices. Then (G_n, o_n) converges locally weakly to $(\mathbb{Z}, 0)$, where o_n is acquired uniformly at random, and $(\mathbb{Z}, 0)$ is the two way infinite path.*

Proof. It is easy to see that if $I = (\mathbb{Z}, 0)$, then $I(r)$ consists of paths that have at most $2r+1$ vertices. What is left to check is whether

$$\frac{1}{n} \sum_{u \in V(G_n)} \mathbb{P}(B_r^{G_n}(u)) \simeq P_{2r+1}) \to \mu(B_r^{\mathbb{Z}}(0) \simeq P_{2r+1})$$

holds. The right-hand side equals to 1, and the left-hand side equals to

$$\frac{1}{n} \sum_{u \in V(G_n)} \frac{n-2r}{n} = \frac{n-2r}{n},$$

which converges to 1. $\square$

Local limit of marked graphs

The definition of local convergence can be adapted for marked graphs as well. This will be important for us in the future, since we look at preferential attachment trees as marked graphs. Let $G(V(G), E(G))$ be a locally finite graph, and $M(G)$ be a mapping that maps G in a complete separable metric space, denoted by Σ. The elements of Σ are called marks, and Σ is called the set of marks. Each edge is given two marks, one associated with each endpoint of an edge in $E(G)$. We denote a marked rooted graph by $(G, o, M(G)) = (V(G), E(G), o, M(G))$. With these at hand we are ready to give the definition of distance on marked graphs.

Definition 2.1.11 *Leg d_Σ be a metric on the space of marks Σ, and let*

$$\begin{aligned} R^\star = \sup\{r : B_r^{G_1} \simeq B_r^{G_2}, & d_\Sigma(m_1(u), m_2(\phi(u)) \leq \frac{1}{r}, \ \forall u \in V(B_r^{G_1}(o_1)), \\ & d_\Sigma(m_1(e), m_2(\phi(e)) \leq \frac{1}{r}, \ \forall e \in E(B_r^{G_1}(o_1))\}, \end{aligned}$$

with $\phi : V(B_r^{G_1}) \to V(B_r^{G_2})$ running over all isomorphisms between $B_r^{G_1}(o_1)$ and $B_r^{G_2}(o_2)$. Then we define the distance of marked graphs as

$$d_{\mathcal{G}_\star}((G_1, o_1, M_1(G_1), G_2, o_2, M_2(G_2)) = \frac{1}{1+R^\star}.$$

With this definition at hand every theorem on local convergence can be stated for marked graphs. Now we move on to discussing the local limit of random graphs.

2.2 Local limit of random graphs

At this point we wish to generalize the notion of weak convergence to random graphs. The difficulty that lies ahead is the following: if we wish to understand $\mathbb{E}(h(G_n, o_n))$, it depends not only on sampling o_n, but also on the underlying random graph structure of G_n. This will result – similarly to the case of the convergence of random variable – in multiple notions of convergence, where the difference between them will be the properties they imply.

Definition 2.2.1 (Local weak convergence of random graphs) *Let $G_n = (V(G_n), E(G_n))$ be a finite random graph. Then we say that G_n converges locally*

- *weakly to $(\overline{G}, \overline{o})$, having law $\overline{\mu}$, when*

$$\mathbb{E}(h(G_n, o_n)) \to \mathbb{E}_{\overline{\mu}}(h(\overline{G}, \overline{o}),$$

  for every bounded continuous function $h : \mathcal{G}_\star \to \mathbb{R}$, where the expectation on the left-hand side is with respect to sampling o_n, and the random graph G_n. We write this as $(G_n, o_n) \xrightarrow{d} (G, o)$.

- *in probability to (G, o), having law μ, when*

$$\mathbb{E}(h(G_n, o_n)|G_n) \xrightarrow{\mathbb{P}} \mathbb{E}_\mu(h(G, o)),$$

  for every bounded continuous function $h : \mathcal{G}_\star \to \mathbb{R}$. We write this as $(G_n, o_n) \xrightarrow{\mathbb{P}-loc.} (G, o)$.

- *almost surely to (G, o), having law μ, when*

$$\mathbb{E}(h(G_n, o_n)|G_n) \xrightarrow{a.s.} \mathbb{E}_\mu(h(G, o)),$$

  for every bounded continuous function $h : \mathcal{G}_\star \to \mathbb{R}$. We write this as $(G_n, o_n) \xrightarrow{a.s-loc.} (G, o)$.

Before we move on from these definitions it is important to emphasize the fact that if we look at the definitions of local convergence in probability, and almost sure local convergence, the conditional expectations on the left-hand side are G_n-measurable. This means that those two conditional expectations are just elementary random variables, and the elementary probabilistic limit theorems apply.
We are going to state a theorem similar to Theorem 2.1.8 for random graphs.

Theorem 2.2.2 ([VDH21] Theorem 2.13) *Let $(G_n)_{n\in\mathbb{N}^+}$ be a sequence of rooted graphs.*

- $(G_n, o_n) \xrightarrow{d} (\overline{G}, \overline{o})$, *having law $\overline{\mu}$ if for all $H_\star \in \mathcal{G}_\star$, and for every $r \geq 0$*

$$\mathbb{E}(p^{G_n}(H_\star)) = \frac{1}{|V(G_n)|} \sum_{u \in V(G_n)} \mathbb{P}(B_r^{G_n}(v) \simeq H_\star) \to \overline{\mu}(B_r^G(\overline{o}) \simeq H_\star), \tag{2.1}$$

- $(G_n, o_n) \xrightarrow{\mathbb{P}-loc} (G, o)$, *having probability law μ, if for every $H_\star \in \mathcal{G}_\star$, and for all integers $r \geq 0$,*

$$p^{G_n}(H_\star) = \frac{1}{|V(G_n)|} \sum_{u \in V(G_n)} \mathbb{1}_{\{B_r^{G_n}(v) \simeq H_\star\}} \xrightarrow{\mathbb{P}} \mu(B_r^G(o) \simeq H_\star).$$

  $(G_n, o_n) \xrightarrow{a.s.-loc.} (G, o)$ *having probability law μ, if for all $H_\star \in \mathcal{G}_\star$ and for all integers $r \geq 0$,*

$$p^{G_n}(H_\star) = \frac{1}{|V(G_n)|} \sum_{u \in V(G_n)} \mathbb{1}_{\{B_r^{G_n}(v) \simeq H_\star\}} \xrightarrow{a.s.} \mu(B_r^G(o) \simeq H_\star).$$

We are not going to get into more details of local limit of random graphs, because we wish to concentrate on the consequences of local weak convergence. The local limit determines many characteristic of a network, such as the neighborhood sizes, the distance of uniformly selected degrees, and others. These and similar theorems can be found in [VDH21] Section 2.4. The reason for this is that if a property of a random graph can be expressed as a continuous, bounded function of the graph, due to the definition of the local limit, the function will have a limit. The problems, however are the following:

- Some properties cannot be expressed as a continuous and bounded function of a graph. This can be solved in some cases, however it complicates things.
- The local limit of a graph is not easily understandable in some cases.

We are not going to investigate the first problem, since it is not relevant for the topic of this thesis, however more can be read in [VDH21] Section 2.4, 2.5. In order to demonstrate the difficulty of the second problem, and due to the importance for us in the next chapter we are going to inspect the local limit of the preferential attachment graph.

2.3 Local limit of the preferential attachment graph

In this section we wish to introduce the local limit of the preferential attachment tree. While doing so, we will define the random tree, that will serve as a limiting object to a modification of the preferential attachment model, with no self loops allowed. The reason for not defining said modification earlier is the fact that it coincides with the definition given in Definition 1.1.1 for trees, and that is the model we wish to further inspect.

2.3.1 The Pólya point graph

The local limit of preferential attachment graphs will be the so-called Pólya point graph. The aim of this section is to give its definition. The definition and corresponding theorems and proofs can be found in section 5.3 of [VDH21]. We start by setting some constants. Let

$$u = \frac{\delta}{2m},$$

$$\chi = \frac{1+2u}{2+2u} = \frac{m+\delta}{2m+\delta},$$

and

$$\psi = \frac{1-\chi}{\chi} = \frac{1}{1+2u} = \frac{m}{m+\delta}.$$

Let $\mathcal{T}$ be a random rooted tree, which is defined by the labeling of its vertices as in the Ulam–Harris labeling of trees. We also assign to each vertex a label y or o depending on whether the age of the vertex is younger or older than of its parent.

The age of vertices are determined recursively as follows. The root $\emptyset$ has an age $U_\emptyset$, where $U_\emptyset$ has uniform distribution on the interval $[0,1]$. In the recursion step we assume that

$$w = \emptyset w_1 w_2 \dots w_l$$

and the corresponding age $x_w \in [0,1]$ has been assigned in a previous step.

For $j \geq 1$, the jth child of w is

$$w = \emptyset w_1 w_2 \dots w_l j,$$

and it is denoted by wj, and let

$$m_-(w) = \begin{cases} m, \text{ if } w \text{ is the root or type } o \\ m-1, \text{ if } w \text{ is of type } y. \end{cases}$$

Let $Y \sim Gamma(m+\delta,1)$, and $Y^\star \sim Gamma(m+\delta+1,1)$, and we take

$$\Gamma_w \sim \begin{cases} Y, \text{ if } w \text{ is the root or type } o \\ Y^\star, \text{ if } w \text{ is of type } y \end{cases}$$

independently from each other. Let $w1, w2, \dots, wm_-(w)$ be the children of w of type o, and let their ages $A_{w1}, A_{w2}, \dots, A_{wm_-(w)}$ be such that

$$A_{wi} = U_{wi}^{(\tau-1)/(\tau-2)} A_w, \text{ where } (U_{wi})_{i=1}^{m_-(w)} \sim \text{i.i.d. Uniform}[0,1], \text{ and } \tau = 3 + \frac{\delta}{m}.$$

Now we need to describe the children of w of type y. In doing so let $(A_{w(m_-(w))+j})_{j\geq 1}$ be the ordered points of a Poisson point process with intensity

$$\rho_w(x) = \Gamma(\tau-1)\frac{\psi x^{1/(\tau-1)-1}}{A_w^{1/(\tau-1)}}, \tag{2.2}$$

on $[A_w, 1]$, and the vertices $((w(m_-(w))+j))_{j\geq 1}$ have type y.

We are finished with the formal definition of the Pólya point graph, yet it is unseen what the connection between this graph and the preferential attachment graph is.

An intuitive explanation is the following. Imagine a modification of the preferential attachment model, such that no self-loops are allowed, and imagine a vertex v at any given point of the process. It can have 2 types of neighbors, the ones that have been added to the graph before, and v used some of its m edges to connect to them – these correspond to the type of points o–, and some other vertices x_i, that used some of their edges to connect to w – these correspond to the type of vertices y. The number of vertices of type y is defined by a Poisson point process with intensity defined in (2.2).

Keeping track of the labeling of the vertices can be done by simply assigning a random variable to each vertex that corresponds to the age of the vertex. Since we are interested in the asymptotic neighborhood of a uniform vertex, the age of the root corresponds to the limit

$$\lim_{n\to\infty} \frac{o_n}{n} \to \text{ Uniform}[0,1],$$

where o_n is the root, and we need to calculate how the children of o_n behave, and the children of their children etc., which is exactly what the recursion above does. Now before we can state a theorem we need to define the modification of the preferential attachment model that was mentioned above.

Definition 2.3.1 (Preferential attachment model with no self loops) *Let $m \geq 1$, and $\delta > -m$. We start from 2 vertices connected by a single edge, and if a new vertex*

v_{n+1} is added, it connects its $j+1$st edges to the vertex v_{n+1} for all $j \in \{0,1,2,\dots,m-1\}$ with probability

$$\mathbb{P}(v_{n+1,j+1} \to v_i | PA_n^{m,\delta}, \textit{ once we added } j \textit{ edges of } v_{n+1} \textit{ to the graph}) = \frac{d_i(n,j)+\delta}{n(2m+\delta)},$$

where $d_i(n,j)$ is the degree of the vertex i after j edges of v_{n+1} have been attached.

This definition coincides with the one given in Definition 1.1.1 with $m = 1$ and $\delta = 0$. We will now state the theorem.

Theorem 2.3.2 ([VDH21] Theorem 5.8) *Let $m \geq 1$, and $\delta > -m$. The preferential attachment model with no self loops allowed converges locally in probability to the Pólya point tree.*

We will not prove this theorem here, the proof can be read in [VDH21]. The proof mostly relies on applying the second moment method to estimating the number of r-neighborhoods.

A question could be whether the same convergence result holds for the other modifications of the preferential attachment model. The case is they do converge to the Pólya point graph, yet only locally weakly. The following theorem states it explicitly. More can be read on this in [VDH21] Chapter 5.3.

Theorem 2.3.3 ([VDH21] 5.25) *Let $m \geq 1$, and $\delta > -m$. The preferential attachment model with self-loops allowed, and the preferential attachment tree without self-loops, starting from two vertices connected by a single edge, with no self-loops allowed converge locally weakly to the Pólya point tree.*

Remark 2.3.4 *From these two theorems it follows that the preferential attachment tree starting from two vertices connected by a single edge, with no self-loops allowed converge locally weakly and locally in probability to the Pólya point graph.*

Remark 2.3.5 *Another remark, which will be exploited later, is that the local weak limit of the preferential attachment tree starting from two vertices connected by a single edge, with no self-loops allowed, and of $PA_1^{1,\delta}$ are identical, and it is the Pólya point graph.*

Chapter 3

Preferential attachment trees with arbitrary seed

In this chapter we wish to investigate whether distinguishing the initial graph of a preferential attachment tree is possible, given that we know the process at a later stage. First we will define what a preferential attachment tree is starting from an arbitrary seed means, then we will introduce a metric that makes distinguishing possible.
During this chapter if not stated otherwise, all proofs and theorems can be found in [BMR15].

In order to be able to discuss the effect of the seed, we need to modify the definition of the preferential attachment graph starting from an arbitrary tree instead of 2 vertices connected by a single edge.

Definition 3.0.1 *Let T be a labeled tree on vertices $0, 1, \dots, |T|$.*

The preferential attachment tree starting from the seed T on n vertices is

$$PA(n|PA(|T|) = T),$$

and we call T the seed of the graph. We will use the notation $PA(n, T)$ for the preferential attachment graph on n vertices starting from the tree T.

This definition shows that a preferential attachment tree from an arbitrary seed should be viewed as a preferential attachment tree conditioned on the event that the model evolves into the seed first, and then its evolution follows the usual preferential dynamics. If we revisit Definition 1.1.2 and Theorem 1.1.3, it can be easily seen that the average degree and maximum degree will converge almost surely regardless of the seed. Surprisingly though it is not true that the seed has no effect at all on the limiting distribution of the maximum degree. This fact will be exploited at measuring the impact of the seeds. We will be inspecting the effect of the seed from two points of view. The first one is from the point of view of the local limit. In this chapter we will prove that the local limit of the preferential attachment model will be the Pólya point graph regardless of the seed.

The importance of this result is that if we take two preferential attachment trees that evolved from different seeds, every bounded local function of neighborhoods of uniformly picked vertices will behave identically. Intuitively, this means that the local neighborhoods of a preferential attachment graph is independent of its seed.

The second approach is inspecting the total variation distance of two trees starting from different seeds. Let T_1, T_2 be two trees, and let

$$\delta(T_1, T_2) = \lim_{n\to\infty} \delta_{TV}(PA(n, T_1), PA(n, T_2)).$$

The authors in [BMR15] showed that if the degree profile of T_1 and T_2 are different, then $\delta(T_1, T_2) > 0$. We are going to state and prove this theorem as well later in this chapter.

The authors in [BMR15] conjectured that $\delta(T_1, T_2)$ is a metric on trees with at least 3 vertices. This was proven by Curien et al. in [CDKM15]. We are going to discuss those results as well in an upcoming chapter, as the methods involved in that paper are not relevant to us as of now.

3.1 The local weak limit of $PA(n, T)$

Theorem 3.1.1 ([BMR15] Theorem 1) *Let T be an arbitrary tree. The local weak limit of $PA(n, T)$ is the Pólya point graph.*

Proof. The outline of the proof is the following: given a seed T, we will make a forest F corresponding to T, and apply the preferential attachment dynamics to the forest F. The trick will be that if F is constructed wisely, according to Theorem 2.3.3, the local weak limit of $PA(n, F)$ will be the Pólya point graph, and the probability that a neighborhood of a uniform vertex in T and F are not identical will be zero. This will yield that the local limit of $PA(n, T)$ is also the Pólya point graph.

Let T be an arbitrary tree, and $\mathcal{T}$ be the Pólya point graph. We wish to exploit equation (2.1), with $G = \mathcal{T}$, and $G_n = PA(n,T)$. If

$$\mathbb{E}(p^{PA(n,T)}(H_\star)) = \frac{1}{|V(G_n)|} \sum_{u \in V(PA(n,T))} \mathbb{P}(B_r^{PA(n,T)}(v) \simeq H_\star) \to \overline{\mu}(B_r^{\mathcal{T}} \simeq H_\star), \tag{3.1}$$

holds, where $p^{PA(n,T)}(H_\star)$ is the probability is that an r-neighborhood of a uniformly chosen vertex of $PA(n,T)$ is isomorphic to $H_\star$. Then $PA(n,T)$ converges locally weakly to the Pólya point graph. In (3.1) $B_r^{PA(n,T)}(v)$ is the r-neighborhood of vertex v, and $B_r^{\mathcal{T}}$ is the r-neighborhood of $\mathcal{T}$, and $\overline{\mu}$ stands for the distribution of $B_r^{\mathcal{T}}$.

Let F be a forest corresponding to T such that to every vertex v in T we associate $d_T(v)$ isolated nodes in F with self loops, where $d_T(v)$ is the degree of v in T. This means that F consists of $2(|T|-1)$ isolated vertices with self loops, and we use the convention that the degree of a vertex with k edges and one self loop is $k+1$.

The evolution of the graph $PA(n,F)$ is defined the same way, as for arbitrary trees, we also mention that we couple the processes $PA(n,T)$ and $PA(n+|T|-2,F)$ the natural way, namely when an edge is added to a vertex v in T, an edge will be also added to the corresponding component of the $d_T(v)$ components of F, and newly added vertices are always coupled.
Our first observation is that the local weak limit of $PA(n+|T|-2,F)$ is the Pólya point graph, since when sampling a random vertex in $PA(n+|T|-2,F)$, the resulting graph has to be restricted to its connected component, and that is a preferential attachment graph starting from one vertex with a self loop, which means that

$$\lim_{n\to\infty} \left(B_r^{PA(n+|T|-2,F)}(k_n(F)) \simeq H_\star \right) = \mathbb{P}\left(B_r^{\mathcal{T}}(\overline{o}) \simeq H_\star \right),$$

where $k_n(F)$ is a uniformly random vertex in $PA(n+|T|-2,F)$. What is left to be shown is that

$$\lim_{n\to\infty} \mathbb{P}\left(B_r^{PA(n+|T|-2,F)}(k_n(F)) \neq B_r^{PA(n,T)}(k_n(T)) \right) = 0,$$

where $k_n(F)$ and $k_n(T)$ are uniform random vertices of F and T coupled in the natural way, namely if $k_n(F)$ is the j-th newly added vertex in $PA(n+|T|-2,F)$, then $k_n(T)$ is the j-th newly added vertex in $PA(n,T)$. The following inequalities hold for every $x>0$,

$$\lim_{n\to\infty}\mathbb{P}\bigg(B_r^{PA(n+|T|-2,F)}(k_n(F))\neq B_r^{PA(n,T)}(k_n(T))\bigg)$$
$$\leq \mathbb{P}(\exists v\in F \text{ s.t. } v\in B_r^{PA(n+|T|-2,F)},k_n(F)\bigg)\leq \mathbb{P}\bigg(\exists v\in F, d_{PA(n+|T|-2,F}(v)<x\bigg)$$
$$+\,\mathbb{P}\bigg(\exists v\in B_r^{PA(n+|T|-2,F)}(k_n(F)), d_{PA(n+|T|-2,F}(v)\geq x\bigg).$$

For the first inequality we exploited the fact that if a vertex of the forest F is in the neighborhood of $k_n(F)$ it will have a self loop, which is not allowed in $PA(n,T)$, therefore the two neighborhoods will be different. During the second we simply decompose the probability to the events whether $d_{PA(n+|T|-2,F}(v)$ is less than x or not.
It is easy to verify that

$$\lim_{n\to\infty}\mathbb{P}\bigg(\exists v\in F, d_{PA(n+|T|-2,F}(v)<x\bigg)=0$$

holds, since the degree of any vertex of F tends to infinity. For the second term we can exploit that it converges to the Pólya point graph, and therefore

$$\mathbb{P}\bigg(\exists v\in B_r^{PA(n+|T|-2,F)}(k_n(F)), d_{PA(n+|T|-2,F}(v)\geq x\bigg)=\mathbb{P}\bigg(\exists v\in B_r^{\mathcal{T}}(\overline{o}),\ \text{s.t. } d_{\mathcal{T}}(v)\geq u)\bigg).$$

What is left to see is that

$$\lim_{u\to\infty}\mathbb{P}\bigg(\exists v\in B_r^{\mathcal{T}}(\overline{o}),\ \text{s.t. } d_{\mathcal{T}}(v)\geq u)\bigg)=0.$$

However this is true, since every neighborhood in $\mathcal{T}(\overline{o})$ can have only finitely many vertices. This is due to the definition of the Pólya point graph, where every vertex has m neighbors of type o and x neighbors of type y, where x is from a Poisson point process, which is an integrable process. This concludes the proof. □

We want to emphasize that during the proof we utilized Note 2.3.5, when we used that the preferential attachment graph with self loops allowed and the preferential attachment tree starting from 2 vertices connected by a single edge both converge locally weakly to the Pólya point graph. This result gives that the seed has no local effect on the graph. An intuitive explanation for this is that a uniform random vertex of the tree will be far away from the seed, and therefore it cannot provide information regarding the seed.

3.2 Total variation distance

Now we will discuss the main result of [BMR15], namely that with the total variation distance different seeds of preferential attachment trees can be detected. First we will define the total variation distance of probability measures.

Definition 3.2.1 *Let P and Q be probability measures on a σ-algebra $\mathcal{F} \subset \Omega$. The total variation distance of P and Q is*

$$\delta_{TV}(P, Q) = \sup_{A \in \mathcal{F}} |P(A) - Q(A)|.$$

From this we can define the asymptotic total variation distance of two preferential attachment trees with different seeds.

Definition 3.2.2 *Let T and S be arbitrary trees, and*

$$\delta(S, T) = \lim_{n \to \infty} \delta_{TV}(PA(n, S), PA(n, T)).$$

In order to understand this definition better it helps to write

$$\delta(S, T) = \lim_{n \to \infty} \delta_{TV}(PA(n, S), PA(n, T)) = \lim_{n \to \infty} \sup_{A \in \mathcal{F}} |\mathbb{P}(PA(n, S) \in A) - \mathbb{P}(PA(n, T) \in A)|.$$

From this it can be read that $\delta(S, T)$ tends to 0 if the limiting probabilities are identical on every $A \in \mathcal{F}$, and that δ is a pseudometric on isomorphism types of trees with at least 3 vertices. Even more is true, namely that δ is a metric on isomorphism types of trees with at least 3 vertices. This theorem will be proven in the next chapter, since for now, we wish to prove a different result, that seems weaker. We will be referring to the list of degrees in a descending order as degree profile.

Theorem 3.2.3 ([BMR15] Theorem 1) *Let S and T be two finite trees with at least 3 vertices. If their degree profiles $\overline{d}(T)$, and $\overline{d}(S)$ are different, then $\delta(S, T) > 0$.*

During the proof we will see that a stronger result holds, namely that the seed has an impact on the maximum degree. We will calculate the cumulative distribution function of the maximum degree of a preferential attachment graph starting from an arbitrary seed.

In order to set the stage for the proof for Theorem 3.2.3, we need to inspect the maximum degree first.

The distribution of the maximum degree starting from an arbitrary seed

First we will note that the limiting random variable of a fixed vertex starting from an arbitrary tree normalized by $\sqrt{n}$ will be $D_i(T)$, which is the natural adjustment to what can be found in Theorem 1.1.5.

Lemma 3.2.4 ([BMR15] Lemma 1) *Let T be a finite tree, and let $U \subset \{1, 2, \ldots, |T|\}$ be a nonempty set of the vertices of T, and let d be the sum of the degrees of U, namely $d = \sum_{i \in U} d_T(i)$. Then as $t \to \infty$ the following holds:*

$$\mathbb{P}\bigg(\sum_{i \in U} D_i(T) > t \bigg) \sim \frac{\Gamma(2|T|-2)}{2^{d-1}\Gamma(|T|-1/2)\Gamma(d)} t^{1-2|T|+2d} \exp(-t^2/4).$$

Lemma 3.2.5 ([BMR15] Lemma 1) *Let T be a finite tree. For every $L > |T|$ there exists a constant $C(L) < \infty$ such that for every $t \geq 1$ we have*

$$\sum_{i=L}^{\infty} \mathbb{P}\Big(D_i(T) > t \Big) < C(L) t^{3-2L} \exp(-t^2/4).$$

The proof of these lemmas can be read in [BMR15]. From these lemmas we may get a useful result for the maximum degree. Let T be a finite tree and $D_{\max}(T)$ the maximum degree of the preferential attachment model starting from T normalized by $\sqrt{n}$.

Theorem 3.2.6 ([BMR15] Corollary 1) *Let T be a finite tree and let $m = |\{i \in \{1, 2, \ldots, |T|\} : d_T(i) = \Delta(T)\}|$, where $\Delta(T)$ is the maximum degree in T. Then the following holds:*

$$\mathbb{P}\Big(D_{\max}(T) > t \Big) \sim m \frac{\Gamma(2|T|-2)}{2^{\Delta(T)-1}\Gamma(|T|-1/2)\Gamma(\Delta(T))} t^{1-2|T|+2\Delta(T)} \exp(-t^2/4).$$

Proof. We are going to exploit that $D_{\max}(T) = \max_{i \geq 1} D_i$ almost surely, and apply the results from the previous lemmas. First we get

$$\mathbb{P}(D_{\max}(T) > t) \leq \sum_{i=1}^{m} \mathbb{P}(D_i(T) > t) + \sum_{i=m+1}^{|T|} \mathbb{P}(D_i(T) > t) + \sum_{i=|T|+1}^{\infty} \mathbb{P}(D_i(T) > t).$$

Now if we apply Lemma 3.2.4 to the first sum on the right-hand side, and exploit the fact that the other two sums are of smaller order we get that

$$\mathbb{P}(D_{\max}(T) > t) \leq \frac{\Gamma(2|T|-2)}{2^{\Delta(T)-1}\Gamma(|T|-1/2)\Gamma(\Delta(T))} t^{1-2|T|+2\Delta(T)} \exp(-t^2/4),$$

for large enough t. We continue by exploiting the lower bound

$$\mathbb{P}(D_{\max}(T) > t) \geq \sum_{i=1}^{m} \mathbb{P}(D_i(T) > t) - \sum_{i=1}^{m}\sum_{j=1}^{m} \mathbb{P}(D_i(T) > t, D_j(T) > t). \tag{3.2}$$

Applying Lemma 3.2.4again with $U = \{i, j\}$ implies that for any $1 \geq i \geq j \geq m$,

$$\sum_{i=1}^{m}\sum_{j=1}^{m} \mathbb{P}(D_i(T) > t, D_j(T) > t) \leq \mathbb{P}(D_i(T)+D_j(T) > 2t) \leq C_{i,j}(T)t^{1-2|T|+4\Delta(T)} \exp(-t^2). \tag{3.3}$$

By exploiting that the exponent on the right-hand side of (3.3) $-t^2$ is smaller by a constant factor than the exponent $-t^2/4$ in Lemma 3.2.4, we get that the the second sum in (3.2) is of smaller order than the first sum, which yields

$$\mathbb{P}(D_{\max}(T) > t) \geq (1 - o(1)) \sum_{i=1}^{m} \mathbb{P}(D_i(T) > t), \tag{3.4}$$

as t tends to infinity. Applying Lemma 3.2.4 to the right-hand side of (3.4) concludes the proof. □

With this at hand we can get to the proof of Theorem 3.2.3.

Proof of Theorem 3.2.3

We will prove this theorem using case analysis. First we will show that the theorem follows if the maximum degrees of the seed are different, then we will generalize this to seeds that have different degree profiles.

Let S and T be two seeds throughout the whole proof.
For the first case we shall assume that $\mathbf{|S|\text{-}\Delta(S)} \neq \mathbf{|T|\text{-}\Delta(T)}$. Without the loss of generality suppose that $|S| - \Delta(S) < |T| - \Delta(T)$. We are going to exploit that any event regarding the maximum degree of a tree can be expressed as an event regarding the tree, therefore for the total variation distance the following holds:

$$\begin{aligned}\delta_{TV}(PA(n, S), PA(n, T)) &\geq \delta_{TV}(\Delta(PA(n, S)), \Delta(PA(n, T))) \\ &\geq |\mathbb{P}(\Delta(PA(n, S)) > t\sqrt{n}) - \mathbb{P}(\Delta(PA(n, T)) > t\sqrt{n})|,\end{aligned}$$

for any $t > 0$ and $n \geq \max\{|S|, |T|\}$. By taking the limit as n tends to infinity we get that

$$\delta(S, T) \geq \sup_{t>0} |\mathbb{P}(D_{\max}(S) > t) - \mathbb{P}(D_{\max}(T) > t)|,$$

and if we apply Theorem 3.2.6 to the right-hand side and exploit that $|S| - \Delta(S) < |T| - \Delta(T)$, we will get that

$$\mathbb{P}(D_{\max}(S) > t) > \mathbb{P}(D_{\max}(T) > t),$$

for large enough t which concludes the proof.

We continue by showing that the theorem holds for the case $|\mathbf{S}| \neq |\mathbf{T}|$. Without loss of generality we suppose that $|S| < |T|$. If $|S| - \Delta(S) \neq |T| - \Delta(T)$, it follows from the previous case, so we may assume that $|S| - \Delta(S) = |T| - \Delta(T)$. With identical manipulations of the total variation distance we can arrive to

$$\delta(S,T) \geq \sup_{t>0} |\mathbb{P}(D_{\max}(T) > t) - \mathbb{P}(D_{\max}(S) > t)|.$$

For the asymptotic behaviour of $\mathbb{P}(D_{\max}(T) > t)$, we can apply Theorem 3.2.6 with $m = 1$, and we need to find an upper bound for $\mathbb{P}(D_{\max}(S) > t)$.
First it can be seen that $\Delta(PA(|T|, S)) \leq \Delta(T)$ with equality holding if and only if every newly added vertex connected to the vertex in S that had the largest degree. This implies that if $\Delta(PA(|T|, S) = \Delta(T)$, then there is exactly one vertex $j \in \{1, 2, \ldots, |T|\}$ such that $d_{PA(|T|,S)}(j) = \Delta(T)$. Applying Theorem 3.2.6 to this we get

$$\mathbb{P}(D_{\max}(S) > t | \Delta(PA(|T|, S)) < \Delta(T)) = o\Big(t^{1-2|T|+2\Delta(T)} \exp(-t^2/4)\Big),$$

as t tends to infinity.

Conditioning on the event $\Delta(PA(|T|, S)) = \Delta(T)$, and exploiting Theorem 3.2.6 again would yield

$$\begin{aligned}&\mathbb{P}(D_{\max}(S) > t | \Delta(PA(|T|, S)) = \Delta(T)) \\ &\leq (1 + o(1)) \frac{\Gamma(2|T| - 2)}{2^{\Delta(T)-1}\Gamma(|T| - 1/2)\Gamma(\Delta(T))} t^{1-2|T|+2\Delta(T)} \exp(-t^2/4),\end{aligned}$$

as t tends to infinity. Combining these we would get

$$\begin{aligned}&\mathbb{P}(D_{\max}(S) > t) \leq (1 + o(1))\mathbb{P}(\Delta(PA(|T|, S)) = \Delta(T)) \\ &\cdot \frac{\Gamma(2|T| - 2)}{2^{\Delta(T)-1}\Gamma(|T| - 1/2)\Gamma(\Delta(T))} t^{1-2|T|+2\Delta(T)} \exp(-t^2/4)\end{aligned}$$

as t tends to infinity. Now we need to combine this with the tail behaviour of $D_{\max}(T)$ which would yield

$$\mathbb{P}(D_{\max}(T) > t) - \mathbb{P}(D_{\max}(S) > t) \geq (1 - o(1))\mathbb{P}(\Delta(PA(|T|, S)) < \Delta(T))) \\ \cdot \frac{\Gamma(2|T| - 2)}{2^{\Delta(T)-1}\Gamma(|T| - 1/2)\Gamma(\Delta(T))} t^{1-2|T|+2\Delta(T)} \exp(-t^2/4)$$

as t tends to infinity. To conclude the proof we need to see that the probability $\mathbb{P}(\Delta(PA(|T|, S)) < \Delta(T)$ is bounded from below. Which holds, since this probability is at least the probability that vertex $|S| + 1$ connects to a leaf of S, which has probability at least $\frac{1}{2|S| - 2}$. This concludes the second case. Combined these we have that if the maximum degree of the seeds S and T differ, then $\delta(S, T) > 0$ holds.

Now we will discuss the case, when **|S|=|T| with different degree profiles**. Let z be the first index in $\{1, 2, \ldots, |T|\}$ such that $d_S(z) \neq d_T(z)$ and without loss of generality suppose that $d_S(z) < s_T(z)$. For the distribution of $D_{\max}(T)$, we have

$$\mathbb{P}(D_{\max}(T) > t) \geq \mathbb{P}(\exists i \in \{1, 2, \ldots, z - 1\} : D_i(T) > t) + \mathbb{P}(D_z(T) > t) \\ - \sum_{i=1}^{z-1} \mathbb{P}(D_z(T) > t, D_i(T) > t),$$

and for $D_{\max}(S)$ we have

$$\mathbb{P}(D_{\max}(S) > t) \leq \mathbb{P}(\exists i \in \{1, 2 \ldots, z - 1\} : D_i(S) > t) + \sum_{i=z}^{\infty} \mathbb{P}(D_i(S) > t).$$

We continue by coupling the evolution of $PA(n, S)$ and $PA(n, T)$ such a way that the degree of vertices $1, 2, \ldots, z - 1$ stays the same. From this we get

$$\mathbb{P}(\exists i \in \{1, 2, \ldots, z - 1\} : D_i(T) > t) = \mathbb{P}(\exists i \in \{1, 2, \ldots, z - 1\} : D_i(S) > t).$$

Combining these equations we get

$$\mathbb{P}(D_{\max}(T) > t) - \mathbb{P}(D_{\max}(S) > t) \\ \geq \mathbb{P}(D_z(T) > t) - \sum_{i=1}^{z-1} \mathbb{P}(D_z(T) > t, D_i(T) > t) - \sum_{i=z}^{\infty} \mathbb{P}(D_i(S) > t).$$

Now by applying Lemma 3.2.4 we get

$$\mathbb{P}(D_z(T) > t) \sim \frac{\Gamma(2|T| - 2)}{2^{d_T(z)-1}\Gamma(|T| - 1/2)\Gamma(d_T(z))} t^{1-2|T|+2d_T(z)} \exp(-t^2/4),$$

and

$$\sum_{i=1}^{z-1}\mathbb{P}(D_z(T)>t, D_i(T)>t)\leq\sum_{i=1}^{z-1}\mathbb{P}(D_z(T)+D_i(T)>2t)$$
$$\leq\sum_{i=1}^{z-1}(1+o(1))\frac{\Gamma(2|T|-2)}{2^{d_T(z)+d_T(i)-1}\Gamma(|T|-1/2)\Gamma(d_T(z)+d_T(i))}t^{1-2|T|+2(d_T(z)+d_T(i))}\exp(-t^2),$$

and

$$\sum_{i=z}^{\infty}\mathbb{P}(D_i(S)>t)\leq Ct^{1-2|T|+2d_S(z)}exp(-t^2/4),$$

for some constant C. By exploiting that $d_S(z)<d_T(z)$, and that $t^x\exp(-t^2)=o(\exp(-t^2/4))$ for any x yields

$$\mathbb{P}(D_{\max}(T)>t)-\mathbb{P}(D_{\max}(S)>t)$$
$$\geq(1-o(1))\frac{\Gamma(2|T|-2)}{2^{d_T(z)-1}\Gamma(|T|-1/2)\Gamma(d_T(z))}t^{1-2|T|+2d_T(z)}\exp(-t^2/4),$$

which combined with

$$\delta(S,T)\geq\sup_{t>0}|\mathbb{P}(D_{\max}(T)>t)-\mathbb{P}(D_{\max}(S)>t)|$$

concludes the proof. □

This result tells us that the initial configuration's effect does not fade as time goes by, if the seeds have different degree profiles. We will prove that δ is a metric on isomorphism types of trees with at least 3 vertices in the next chapter.

3.3 Some notes regarding $\delta(S,T)$ and percolation

In [BMR15] there are several open problems mentioned. Two of them are how could this question be formed into either hypothesis testing, or how could the initial seed be estimated from the preferential attachment tree at a later stage. We were trying to understand how does $\delta(S,T)$ react when the preferential attachment trees can only be observed with some error. It goes without saying that if the statistical background of a problem is examined these questions are of key importance

In this section we will be discussing how $\delta(S,T)$ reacts to percolation of a graph, and as a corollary we prove that the limit of the total variation distance of a preferential and a uniform attachment tree is 1.

Percolation on the preferential attachment graph

We will be discussing how percolation on the preferential attachment graph affects the influence of the seed. First we will be discussing these problems such that we assume that one graph can be observed without any error, while in the other one every edge is omitted independently from each other with probability ε.

After this we will be investigating the case when every edge is omitted from the graph with probability ε, and after that some edges are added to the graph so the result will be a connected tree again.

We are now going to state these propositions explicitly.

Definition 3.3.1 *Let ε be in $[0,1]$, and G be an arbitrary graph, and G^ε be the graph, such that*

$$V(G) = V(G^\varepsilon),$$

and for every edge $e \in E(G)$ the following holds

$$\mathbb{P}(e \in E(G^\varepsilon)) = 1 - \varepsilon,$$

independently from every other edge, and no edge is added to the graph.

We will use the following notation for the liminf of the percolation of a preferential attachment graph starting from a seed:

$$\delta_{\liminf}(S, T^\varepsilon) = \liminf_{n\to\infty} \delta_{TV}(PA(n,S), PA(n,T)^\varepsilon),$$

and use the notation

$$\delta(S, T^\varepsilon) = \lim_{n\to\infty} \delta_{TV}(PA(n,S), PA(n,T)^\varepsilon),$$

if the limit exists.

Before stating the theorem we will have to define another quantity from information theory which is related to the total variation distance.

Definition 3.3.2 (Hellinger Distance) *Let $P = (p_1, p_2, \ldots, p_n)$ and $Q = (q_1, q_2, \ldots, q_n)$ be two finitely supported probability measures. The Hellinger distance of these two probability measures is*

$$H(P,Q) = \frac{1}{\sqrt{2}}\sqrt{\sum_{i=1}^{n}(\sqrt{p_i} - \sqrt{q_i})^2}.$$

Now we will need a lemma that connects the Hellinger distance to the total variation distance. The proof of this can be found in [Har13]

Lemma 3.3.3 ([Har13]) *Let P and Q be two discrete probability measures. The following holds*

$$H^2(P,Q) \le \delta_{TV}(P,Q) \le \sqrt{2}H(P,Q),$$

where

$$H^2(P,Q) = \left(\frac{1}{\sqrt{2}}\sqrt{\sum_{i=1}^{n}(\sqrt{p_i}-\sqrt{q_i})^2}\right)^2,$$

and

$$1 - H^2(P,Q) = \sum_{i=1}^{k}\sqrt{p_i q_i}.$$

Proposition 3.3.4 *Let T and S be two arbitrary trees, and $PA(n,S)$, and $PA(n,T)$ are the graphs that evolved from the seeds S and T according to the preferential attachment dynamics. Then*

$$\delta(S,T^\varepsilon) = \lim_{n\to\infty} \delta_{TV}(PA(n,S), PA(n,T)^\varepsilon)$$

exists, and

$$\lim_{n\to\infty} \delta_{TV}(PA(n,S), PA(n,T)^\varepsilon) = \liminf_{n\to\infty} \delta_{TV}(PA(n,S), PA(n,T)^\varepsilon) = 1.$$

Proof. Recall that according to Lemma 3.3.3 with $\delta_{TV}(PA(n,S), PA(n,T)^\varepsilon)$ we have that

$$\delta_{TV}(PA(n,S), PA(n,T^\varepsilon)) \ge H^2(PA(n,S), PA(n,T^\varepsilon)).$$

By rearranging the second statement of Lemma 3.3.3, we get that

$$H^2(PA(n,S), PA(n,T^\varepsilon) = 1 - \sum_{\omega\in\Omega}\sqrt{\mathbb{P}(PA(n,S)=\omega)\mathbb{P}(PA(n,T)^\varepsilon=\omega)},$$

where Ω is the set of labeled graphs on n vertices. By decomposing Ω to connected and disconnected graphs, we get

$$1 - \sum_{\omega \in \Omega} \sqrt{\mathbb{P}(PA(n,S) = \omega)\mathbb{P}(PA(n,T)^{\varepsilon} = \omega)} =$$
$$1 - \sum_{\omega \in \Omega,\ \omega \text{ is connected}} \sqrt{\mathbb{P}(PA(n,S) = \omega)\mathbb{P}(PA(n,T)^{\varepsilon} = \omega)}$$
$$- \sum_{\omega \in \Omega,\ \omega \text{ is disconnected}} \sqrt{\mathbb{P}(PA(n,S) = \omega)\mathbb{P}(PA(n,T)^{\varepsilon} = \omega)},$$

which concludes the proof, by exploiting that every element of the sums equals to zero, since $PA(n,S)$ is connected and $PA(n,T)^{\varepsilon}$ is disconnected. □

From this it can be easily seen that if G and $G^{\star}$ are connected and disconnected graphs, then their total variation distance will be 1. This means that even if 1 edge is omitted from a preferential attachment model, while the other one can be observed without any error, the limit of their total variation distance will be 1.

We are going to state this proposition explicitly.

Proposition 3.3.5 *Let T and S be arbitrary (not necessarily different trees), and let $PA(n,T)$ be an arbitrary preferential attachment graph and $PA(n,S)^{\star}$ be a graph such that we omit at least one edge from $PA(n,S)$ for every n. Then*

$$\lim_{n \to \infty} \delta_{TV}(PA(n,T), PA(n,S)^{\star}) = 1.$$

This implies that if a statistical test was to be designed, it has to observe the graphs without any error.

The next problem we investigated was related to this: how could the disconnected graph be connected again. Let G be an arbitrary graph, and G^{ε} as before. We create $\overline{G}^{\varepsilon}$ as follows:

- Shrinking every component of G^{ε} down to a vertex, and making it a complete graph. Let the resulting graph be G°.
- Taking a uniform random spanning tree in G°.
- For every edge uv in G° we sample a vertex x of G^{ε}, which is in the component corresponding to u, and sample another vertex y of G^{ε}, which is in the component corresponding to v. We connect x and y by a single edge.

Doing this will result in a connected graph $\overline{G}^{\varepsilon}$. We will use the following notation:

$$\delta(S, \overline{T}^{\varepsilon}) = \lim_{n\to\infty} \delta_{TV}(PA(n,S), \overline{PA(n,T)}^{\varepsilon}).$$

With this notation at hand we can state our proposition.

Proposition 3.3.6 *Let T and S be arbitrary trees, and $\varepsilon \in (0,1)$, then*

$$\delta(S, \overline{T}^{\varepsilon}) > 0.$$

The proof will follow directly from two lemmas, which we are going to state now.

Lemma 3.3.7 ([DVDHH10] Theorem 1.3) *Fix $m \geq 1$, and $\delta > -m$, then there exists a constant $c(m,\delta) > 0$, such that the diameter of $PA^{m,\delta}(n)$ is at most $c(m,\delta)\log(n)$ with high probability.*

Lemma 3.3.8 ([ANS21] Theorem 1) *For any $\varepsilon, \delta \in (0,1)$, there exists $C(\varepsilon,\delta) \in (1,\infty)$, such that if G is a connected simple graph on n vertices with minimal degree at least δn, then*

$$\mathbb{P}\left(\frac{\sqrt{n}}{C(\varepsilon,\delta)} \leq diam(UST(G)) \leq C(\varepsilon,\delta)\sqrt{n}\right) \geq 1-\varepsilon,$$

where $UST(G)$ is a uniform random spanning tree of G, and diam is its diameter.

Now we can prove our proposition.

Proof of Proposition 3.3.6.

The proof follows by directly applying these two lemmas to $PA(n,S)$, which has diameter at most $c\log(t)$, and to $\overline{PA(n,T)}^{\varepsilon}$, which has diameter $o(\sqrt{n\varepsilon})$, with probability arbitrarily close to 1. By taking the event $diam(G) < n^{0.25}$, for both graphs, and applying the definition of the total variation distance, we get that the limit of the total variation distance remains bounded away from zero. This concludes the proof. □

From these previous lemmas we can also get that if T_n is a uniform attachment tree on n vertices, and P_n is a preferential attachment tree on n vertices, then

$$\liminf_{n\to\infty} \delta_{TV}(P_n, T_n) = 1,$$

and therefore the limit is defined, and

$$\lim_{n\to\infty} (P_n, T_n) = 1.$$

Chapter 4

Decorated trees and the uniform attachment model

In this chapter we are going to prove two results that can be viewed as an extension to [BMR15]. First we will prove the result of Curien et al., that the δ function introduced in the previous chapter is a metric on trees with at least 3 vertices. The proof of said theorem uses the notion of decorated trees, which will be defined later. This method turned out to be powerful enough for Bubeck et al. to investigate the influence of the seed in uniform attachment trees in [BEMR17]. They found that the modification of δ for uniform attachment trees will also be a metric on trees with at least 3 vertices.

4.1 Decorated trees

First we will set the stage for stating and later proving our theorems. In order to do so we will need to define the main tool of this chapter, decorated trees.

Definition 4.1.1 *A decorated tree is a pair $\tau = (\overline{\tau}, \ell)$ where $\overline{\tau}$ is a tree, and ℓ is a family of positive integers, such that $\ell(u)$ is positive for all $u \in \overline{\tau}$. We denote by $|\tau|$ the number of vertices in $\overline{\tau}$, and set*

$$w(\tau) = \sum_{u \in \overline{\tau}} \ell(u),$$

to be the total weight of τ. The set of all decorated trees τ is denoted by $\mathfrak{D}$.

Next we are going to introduce a partial ordering on $\mathfrak{D}$.

Definition 4.1.2 *Let τ, τ' be two decorated trees in $\mathfrak{D}$. We write $\tau \prec \tau'$ if*

$$w(\tau) = w(\tau') \text{ and } |\tau| < |\tau'|$$

or

$$w(\tau) < w(\tau') \text{ and } |\tau| \leq \tau'$$

holds. Thus $\prec$ is a strict partial ordering on $\mathfrak{D}$, we denote by $\preceq$ the associated partial order.

Next we will define what we wish to observe when distinguishing the seeds of preferential attachment network. Let k, j be arbitrary positive natural numbers, and

$$[k]_j = \prod_{i=0}^{j-1}(k-i),$$

and τ, T two trees and $\phi : \tau \to T$ be an embedding. For a decorated tree $\tau = (\overline{\tau}, l)$ let

$$D_\tau(T) = \sum_{\phi \text{ over all embeddings } \tau \to T} \prod_{u \in \overline{\tau}} [\deg_T(\phi(u))]_{\ell(u)},$$

where $\deg_T(x)$ is the degree of vertex x in T. We wish to emphasize that embeddings are defined for trees, and the definition above is defined for a decorated tree τ, and an arbitrary tree T. The fact that τ is a decorated tree is exploited when we use the weights of vertices in the product. It can be easily seen that if τ_1 is a decorated tree formed of a single vertex with label one, then $D_{\tau_1}(T)$ is the total degree of T.

4.2 δ is a metric on trees with at least 3 vertices

With the definition of decorated trees we can state the main lemma that will exploit.

Lemma 4.2.1 ([CDKM15] Proposition 5) *Let τ be a decorated tree. There exist constants*

$$\{c_n(\tau, \tau') : \tau' \preceq \tau, n \geq 2\} \text{ with } c_n(\tau, \tau') > 0,$$

such that for every seed S, for which $|S| = n_0$ the process $(M_\tau^S(n))_{n \geq n_0}$ defined by

$$M_\tau^S(n) = \sum_{\tau' \preceq \tau} c_n(\tau, \tau') D_{\tau'}(PA(n, S))$$

is a martingale with respect to the filtration $\mathcal{F}_n = \sigma(PA(n_0, S), PA(n_0+1, S), \ldots, PA(n, S))$.

The proof of this lemma is similar in nature to the proof of Theorem 1.1.2, as first the conditional expectations are calculated with respect to the filtration, then certain asymptotic behaviours are calculated for the conditional expectations, and lastly the martingales are constructed. The asymptotic behaviour can be acquired by exploiting that the number of embeddings can be calculated with combinatorial reasoning. The proof can be found in [CDKM15], but now we wish to state the main theorem of this section.

Theorem 4.2.2 (Theorem 1 [CDKM15]) *The function δ as introduced in the previous chapter is a metric on trees with at least* 3 *vertices.*

Proof. We wish to emphasize that the only thing left to verify is that if $S \neq T$ then $\delta(S,T) \neq 0$ holds. The proof mostly relies on Lemma 4.2.1. Let $S \neq T$ be two distinct trees on 3 vertices. We claim that if $n_0 = \max(|S|, |T|)$, then there exists a deterministic decorated tree τ such that

$$\mathbb{E}(D_\tau(PA(n_0, S)) \neq \mathbb{E}(D_\tau(PA(n_0, T)). \tag{4.1}$$

To see this first we can assume that $|S| \leq |T|$, and set $S' = PA(|T|, S)$. This means that if $|S| < |T|$ then S' is a random tree. If we take $\tau = T$ with labels $\ell(u) = \deg(u)$, then for every tree X with $|X| = |T|$ we have that

$$D_\tau(X) = D_\tau(T)\mathbb{1}_{\{T=X\}}.$$

Applying this to trees S' and T, and taking expectations yields

$$\mathbb{E}(D_\tau(S')) = D_\tau(T)\mathbb{P}(S' = T).$$

If $|S| = |T|$ the above probability is 0, and when $|S| < |T|$ it can be easily seen that S' is non-deterministic. The fact that $|S| \geq 3$ is exploited here, and this yields that the probability above is strictly less than 1. In both cases (4.1) holds.

We continue by setting τ to be the minimal decorated tree with respect to the partial ordering $\preceq$ defined in Definition 4.1.2 for which (4.1) holds. Then

$$\mathbb{E}(D_{\tau'}(PA(n_0, S)) \neq \mathbb{E}(D_{\tau'}(PA(n_0, T)) \tag{4.2}$$

holds for all $\tau' \prec \tau$ and it follows that

$$\mathbb{E}M_\tau(PA(n_0, S) \neq \mathbb{E}M_\tau(PA(n_0, T)),$$

where martingales $M_\tau(PA(n_0, S)$ and $M_\tau(PA(n_0, T)$ are defined in Lemma 4.2.1. We are going to use these martingales to bound the total variation distance. For $n \geq n_0$

$$\delta_{TV}(PA(n,S), PA(n,T)) \geq \delta_{TV}(M_\tau(PA(n,S)), M_\tau(PA(n,T)))$$
$$\geq \frac{(\mathbb{E}[M_\tau(PA(n,S)) - M_\tau(PA(n,T))])^2}{2(Var(M_\tau(PA(n,S))) + Var(M_\tau(PA(n,T)))) + (E[M_\tau(PA(n,S)) - M_\tau(PA(n,T))])^2}$$

holds, since

$$\mathbb{E}[M_\tau(PA(n,S)) - M_\tau(PA(n,T))] = \mathbb{E}[M_\tau(PA(n_0,S)) - M_\tau(PA(n_0,T))] \neq 0,$$

and

$$Var(M_\tau(PA(n,S))) + Var(M_\tau(PA(n,T)))$$

is bounded as n tends to infinity, since the two martingales are bounded in L^2. This yields that $\delta_{TV}(S,T)$ is uniformly bounded away from 0 as n tends to infinity, which concludes the proof. $\square$

Now we can turn to investigating the uniform attachment model. We will not go into such depths as with the preferential attachment model, we only want to emphasize that the methods of this section can be applied to a different class of random trees.

4.3 The uniform attachment model

In this section we will present the analogous version of Theorem 4.2.2. In order to do so we are going to adjust the definitions involved to the uniform attachment tree. First we give a short introduction to the uniform attachment tree, and then state some theorems on the asymptotic behaviour of the uniform attachment tree.

Definition 4.3.1 *We start from a simple vertex, and at every step a new vertex is added to the graph, which connects uniformly at random to one of the already existing vertices. Similarly to $PA(n)$, we denote the uniform attachment model after n steps by $UA(n)$.*

4.3.1 Some results regarding the unifrom attachment tree

Now we wish to collect few theorems regarding some properties of the uniform attachment tree. We are going to state theorems that can be found in [Drm09] with their proofs. Our first result is regarding the maximum degree of the uniform attachment model.

Theorem 4.3.2 (Maximum degree [Drm09] Theorem 6.12) *Let $\Delta(UA(n))$ denote the maximum degree of the uniform attachment tree after n steps. Then the following holds*

$$\mathbb{P}(\Delta(UA(n)) \leq d) = \exp\bigg(- 2^{-(d-\log_2 n+1)}\bigg) + o(1).$$

Another result is regarding the height of the uniform attachment tree.

Theorem 4.3.3 (Height of the uniform attachment tree [Drm09] Theorem 6.32) *Let H_n denote the height of the uniform attachment tree after n steps. Then*

$$\mathbb{E}(H_n) \sim e \log n,$$

and

$$\mathbb{P}\bigg(|H_n - \mathbb{E}H_n| \geq \eta\bigg) = O(e^{-c\eta})$$

with some constant c.

This also means that the variance of the height of the uniform attachment tree tends to zero.

4.3.2 The seed of the uniform attachment tree

The question remains the same: does conditioning on the event that $UA(k) = S$ for some tree S have an effect on $UA(n)$ if $n > k$. In order to study this we need to define how the evolution of a tree should be defined from an arbitrary seed. Let S be an arbitrary graph on k vertices, and we define $UA(n, S)$ by induction. First $UA(k, S) = S$, then given $UA(m, S)$ we acquire $UA(m+1, S)$, by adding a new vertex v to the graph and sampling a vertex u uniformly at random from the m already existing vertices, then adding the edge uv to the graph.

We will use the same definition for δ as before, but in order to emphasize that we apply it for the uniform attachment model we will use the notation δ^U for it.

$$\delta^U(S, T) = \lim_{n\to\infty} \delta_{TV}(UA(n, S), UA(n, T)).$$

This limit is always well defined, since the total variation is bounded, and it is decreasing in n.

Theorem 4.3.4 (Theorem 1 [BEMR17]) *For any S, T non-isomorphic trees with at least 3 vertices*

$$\delta^U(S, T) > 0.$$

The proof can be found in [BEMR17]. The proof is similar to what we did in Theorem 4.2.2, as we can state a lemma similar to Lemma 4.2.1, and the theorem follows from that. The main difference between the two results regarding the preferential and the uniform attachment model lies in the proofs of the two lemmas. The reason for that is the different characteristics of the graph sequences, and therefore the different approach to calculating the coefficients of the martingales.

Chapter 5

Summary and further directions

In this final chapter we wish to summarize the main results of this thesis, and mention few possible ways to continue with this topic.
The first two chapters were mainly focused on introducing the variants of the preferential attachment model, and the local weak limit. The third chapter was about the total variation distance of two preferential attachment graph sequences and their properties. Our results showed that the total variation distance of preferential attachment graphs is sensitive to error, therefore if a test was to be designed, the whole graph had to be observed without any error. This coincides with the result regarding the local weak limit of the preferential attachment graphs, namely that the seed has no effect locally on the graph. In the fourth chapter we proved that the function introduced in [BMR15] is a metric indeed. We also stated that the same results hold for uniform attachment trees.

Some open questions would be how the different variations of the preferential attachment model react to the change of their seed. It can be shown that the maximum and average degree of the variants of the preferential attachment model with additive and multiplicative fitness behave asymptotically similarly to the basic preferential attachment model. Also in 2021 in [Lo21](Theorem 1.5.) it was shown that under some assumptions regarding the fitness, the local weak limit of the preferential attachment tree with additive fitness is the Pólya point tree. These indicate that similar results might be true for models with additive fitness as well.
Another interesting problem could be to design an algorithm to find the seed of the preferential attachment tree. There are some known results for the uniform attachment tree in [LP19], which we are going to state now.

5.1 Algorithmic results for the uniform attachment tree

The first two results state that if the seed is of a special type, a great amount of vertices of it can be acquired.

Theorem 5.1.1 ([LP19] Theorem 1.1) *Let* $\varepsilon \in (0, 1)$, $\gamma \in (0, 1)$ *and let*

$$\ell \geq \max \left\{ \frac{2e^2}{\gamma} \log\left(\frac{1}{\varepsilon}\right), \frac{2e^2}{\gamma} \log(4e^2) \right\}$$

be a positive integer. Then for all $n \geq \ell$ *sufficiently large if* $UA(n, P_\ell)$ *is a uniform attachment tree with root* P_ℓ*, where* P_ℓ *is a path of* ℓ *vertices, then there exists a path finding algorithm that outputs a vertex set* $H_n \subset \{1, 2, \ldots, n\}$*, with* $|H_n| \geq (1 - \gamma)\ell$ *such that*

$$\mathbb{P}(H_n \subset P_\ell) \geq 1 - \varepsilon$$

holds.

The following theorem states that if the seed is a star, a set of vertices can be found algorithmically, which contains the seed with high probability.

Theorem 5.1.2 ([LP19] Theorem 1.3) *There exists a positive constant* C *such that the following holds. Let* $\varepsilon \in (0, 1)$, $\gamma \in (0, 1)$*, and*

$$\ell \geq \max(C, 8/\gamma) \log\left(\frac{1}{\varepsilon}\right)$$

be a positive integer. Then for all $n \geq \ell$ *sufficiently large if* $UA(n, S_\ell)$ *is a uniform attachment tree with seed* S_ℓ*, such that* S_ℓ *is a star on* ℓ *vertices, then there exists a seed finding algorithm that outputs a vertex set* H_n*, with* $|H_n| \leq (1 + \gamma)l$ *such that*

$$\mathbb{P}(S_\ell \subset H_n) \geq 1 - \varepsilon.$$

The last result we wish to state from [LP19] is that regardless of the type of the seed, there is an algorithm that can find a set of vertices of it.

Theorem 5.1.3 ([LP19] Theorem 1.5) *Let* $UA(n, T_\ell)$ *be a uniform attachment tree on* n *vertices starting from seed* T_ℓ*, and let* $\varepsilon > 0$*, and* $\ell \geq 1$*, and let*

$$a = 2 \log\left(\frac{4\ell^2}{\varepsilon}\right) + 1.$$

If

$$\ell \geq 64a^2 \log\left(\frac{22a\ell^2}{\varepsilon}\right)$$

holds for ℓ, then there exists a seed finding algorithm that outputs a vertex set $H_n \subset \{1, 2, \ldots, n\}$ with $|H_n| \geq \ell/3a$, such that

$$\liminf_{n\to\infty} \mathbb{P}(H_n \subset T_\ell) \geq 1 - \varepsilon.$$

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Influence of initial configuration in growing random graph models

A **diplomamunka** szerzőjeként fegyelmi felelősségem tudatában kijelentem, hogy a dolgozatom önálló szellemi alkotásom, abban a hivatkozások és idézések standard szabályait következetesen alkalmaztam, mások által írt részeket a megfelelő idézés nélkül nem használtam fel.

Budapest, 2021.12.30.

a hallgató aláírása